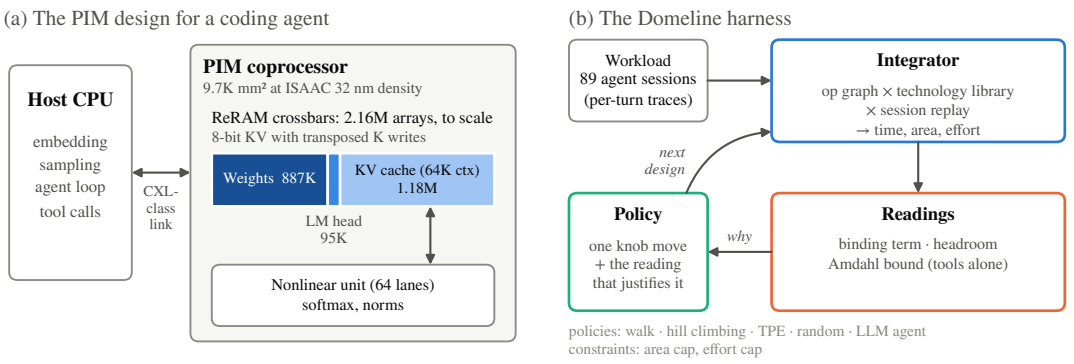
(a) The PIM design for a coding agent
Host CPU
embedding
sampling
agent loop
tool calls
CXL-class link
PIM coprocessor
9.7K mm² at ISAAC 32 nm density
ReRAM crossbars: 2.16M arrays, to scale
8-bit KV with transposed K writes
Weights 887K
KV cache (64K ctx) 1.18M
LM head 95K
Nonlinear unit (64 lanes)
softmax, norms
(b) The Domeline harness
Workload
89 agent sessions
(per-turn traces)
Integrator
op graph × technology library
× session replay
→ time, area, effort
next design
Policy
one knob move
+ the reading
that justifies it
why
Readings
binding term · headroom
Amdahl bound (tools alone)
policies: walk · hill climbing · TPE · random · LLM agent
constraints: area cap, effort cap